

Machine Learning

DD2421, 7.5 credits

Atsuto Maki, André Pereira, Jörg Conradt

Autumn 2026

1 About the course

- Course Contents
- Textbooks
- Who are teaching?

2 Logistics

- Lectures
- Labs
- Examinations
- Miscellaneous

3 A very brief overview of Machine Learning

- Applications
- Types of Learning
- Supervised and Unsupervised

The aim of the course is to provide:

- basic knowledge of the most important algorithms and theory that form the foundation of machine learning
- a practical knowledge of machine learning algorithms and methods

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Course contents:

- Lectures 1–11
- Lecture 12, mini lectures + exam Q&A
- LAB1 (labs 1–3, **NB. there is a deadline for each**)
- Written exam

Intended outcomes – students will be able to:

- describe the most important algorithms and the theory that constitutes the basis for machine learning and artificial intelligence
- explain the principle for machine learning and how the algorithms and the methods can be used
- discuss advantages with and limitations of machine learning for different applications

in order to be able to identify and apply appropriate machine learning technique for classification, pattern recognition, regression and decision problems.

Recommended reading

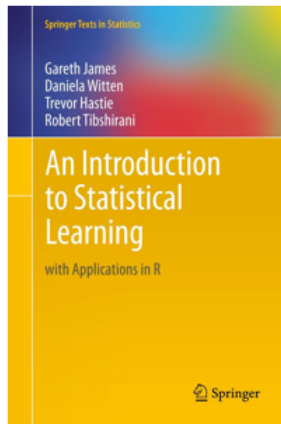
Gareth James, Daniela Witten,
Trevor Hastie and Robert Tibshirani

An Introduction to Statistical Learning

Springer, 2021

Available online:

<https://faculty.marshall.usc.edu/gareth-james/ISL/>



Recommended reading

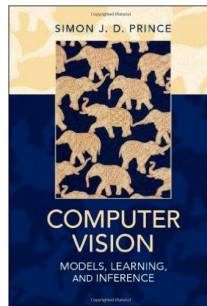
Simon Prince

Computer Vision: Models, Learning,
and Inference

Cambridge University Press, 2012

Available online:

web4.cs.ucl.ac.uk/staff/s.prince/book/book.pdf



NB. Pointers to other reference materials under Lectures page.

Who are teaching?

- **Atsuto Maki**
Dept. Robotics, Perception, and Learning (RPL)
- **André Tiago Abelho Pereira**
Dept. Speech, Music, and Hearing (TMH)
- **Jörg Conradt**
Dept. Computational Science and Technology (CST)
- Course Assistant: **Alexander Kozlov**
Dept. Computational Science and Technology
- ca. 10 teaching assistants (the majority is PhD students)

André Tiago Abelho Pereira

*From simple transactional,
disembodied interactions*

Research Mission



*To complex **social**
embodied interactions*



Research topics: Human-Robot Interaction, Multimodal Machine Learning

Example Projects:

*fMRI as a modality to study
conversational engagement
with humans and robots*



*multimodal LLMs for
collaborative embodied AI
in games and education*



*robot telepresence for data
collection and multimodal
model creation*

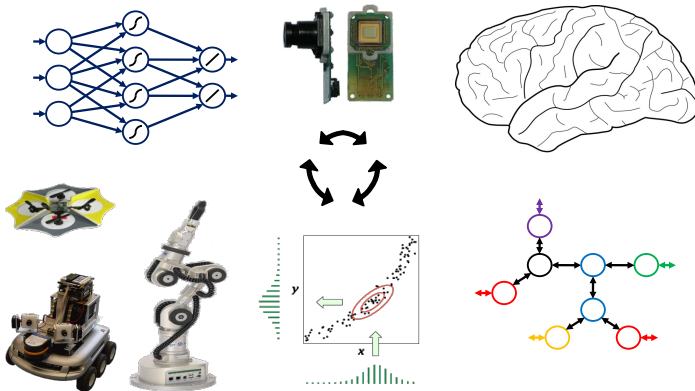


Teaching

- Artificial Intelligence (Course responsible)
- Topics in Social Robotics (PhD Course)
- Machine Learning (this course)

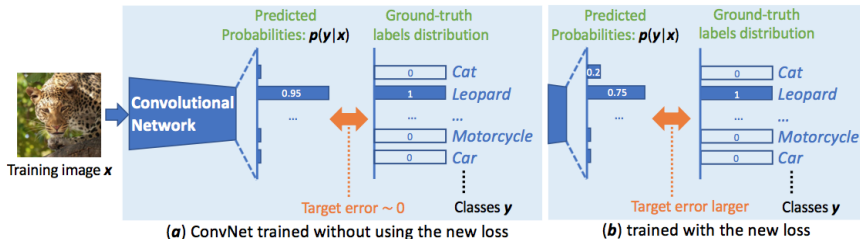
Jörg Conradt

Research Topic: Neural Computation for Real-Time System Engineering



Atsuto Maki

Research topics: Computer Vision, Machine/Deep Learning



<https://robotics.sciencemag.org/content/4/30/eaaw1329.full>

<https://www.kth.se/profile/atsuto>

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Course Information on Canvas

<https://canvas.kth.se/courses/63976>

Course Information on KTH Social

<https://www.kth.se/social/course/DD2421/>

<https://www.kth.se/social/course/DD2421/calendar/>

<https://www.kth.se/social/course/DD3431/>

Course registration needed!

Any inquiries to be directed to Student service support EECS.

For administrative questions please consult this page:

www.kth.se/en/eecs/studentsupport

Lectures

- 1 Nearest Neighbour Classifier (Memory-based)
- 2 Decision Trees (Logical inference)
- 3 Challenges in Machine Learning
- 4 Regression
- 5 Support Vector Machines
- 6 Artificial Neural Networks
- 7 Probabilistic Methods
- 8 Learning as Inference
- 9 Learning with Latent Variables
- 10 Ensemble Methods
- 11 Dimensionality Reduction
- 12 Mini lectures (beyond the scope of DD2421), exam Q&A

LAB1 (3.5 credits)

- ① Decision Trees
 - ② Support Vector Machines
 - ③ Bayes Classifier & Boosting
- labs are carried out by students and examined by TAs
 - use Canvas to book time slots **before the deadlines**

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Examination:

- It is **your** task to convince the examiner that you have done the assignment and understood the results.
- Strongly encouraged to work+present by pairs of 2 students (not 3!).
- 10 minutes, **be there on time**.
- No programming code to be shown
- Bring your ID (tell the TA if you are yet to be registered)

Examination (4 credits)

A written examination.

Date: Fri 16 October 14:00-18:00

Exam Registration in advance!

<https://www.kth.se/form/exams>

Registration opens 15 September and closes 29 September
at 23.59. (please double-check!)

Set a reminder by yourself! (teachers cannot help if you missed it)

Chance for re-exam (16 December 14:00-18:00), registration required.

FAQ

Q. Are course slides available?

A. Will be uploaded on Canvas under the "Lectures" in advance.

Q. Can you register me to the course / examination?

A. No, please consult Student service support EECS:
www.kth.se/en/eecs/studentsupport

PhD students

Please inform/specify your KTH-account by email with subject “FDD3431 HT26” addressed to the examiner.

Furhter instructions for extra report will be provided.

If you are from another Swedish university, a form to get a KTH-account available at the reception of EECS.

See the instructions:

<https://intra.kth.se/en/eecs/forskarutbildning/courses>

Miscellaneous

- * Message board available on “Discussion” on KTH Canvas (but bear with us – teachers cannot promise to respond :-)).
- * Course committee: It will be a great pleasure to have students’ course committee (a.k.a. kursnämnd). Anyone volunteers, please?

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Sample Applications

- Image recognition / Computer vision
- Speech recognition and synthesis
- Natural language processing
- Autonomous robots
- Spam-filter for e-mail
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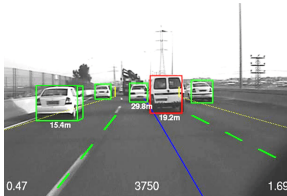
Where is machine learning useful?

A pattern exists

Data available for training

Hard/impossible to define rules mathematically

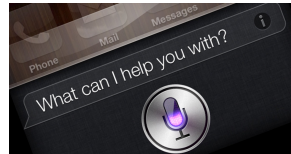
Driving assistants (Google, Toyota, Volvo, ...)



Board games (DeepMind AlphaGo, ...)



Personal assistants (Apple Siri, Amazon Eco, ...)



LLM, Translation (OpenAI GPT, ...)



Types of Learning

- **Supervised Learning** (covered)
- **Unsupervised Learning** (briefly covered)
- **Reinforcement Learning** (not covered)
- **Evolutionary Learning** (not covered)

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 - Consider how a baby learns to walk for instance.
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 - General Purpose Optimization

Supervised vs Unsupervised learning paradigms

Conflates two different distinctions:

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 - Learning mappings from A to B.
(Neutral mathematics.)
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- **Unsupervised Learning**, a.k.a. descriptive
 - Analyzing unstructured raw data. There is no B, only A.
 - Learning without human supervision.
(Scalable and biologically plausible.)